MaxAB Digital Wallet Analysis Project

Comprehensive Analytical Challenge Solution

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Question 1: Collection Orders Analysis

Analyzing weekly success rates, agent speed, and regional productivity for cash collections[cite: 11, 12].

Part A: Weekly Success Rates

```
SELECT DATEPART(week, collection_date) AS [Week], COUNT(*) AS
Total, COUNT(CASE WHEN ACTION IN ('COLLECTED',
'PARTIALLY_COLLECTED') THEN 1 END) AS Success,
ROUND(CAST(COUNT(CASE WHEN ACTION IN ('COLLECTED',
'PARTIALLY_COLLECTED') THEN 1 END) AS FLOAT) / CAST(COUNT(*) AS
FLOAT) * 100, 2) AS Success_Rate FROM Collection_orders GROUP BY
DATEPART(week, collection_date);
```

Finding: A success rate is defined as orders either collected or partially collected[cite: 14]. This metric monitors the stability of field operations over time.

Part B: Agent Completion Speed (April)

```
-- Ranking agents from Start to Finish for April [cite: 15] SELECT a.agent_id, AVG(DATEDIFF(minute, a.action_timestamp, b.action_timestamp)) AS Avg_Minutes FROM Collection_orders a JOIN Collection_orders b ON a.collection_order_id = b.collection_order_id WHERE a.ACTION = 'STARTED TASK' AND b.ACTION IN ('COLLECTED', 'FAILED') AND MONTH(a.collection_date) = 4 GROUP BY a.agent_id ORDER BY Avg_Minutes ASC;
```

Part C: Zone Productivity

Zone Name	Productivity (Orders/Agent)	Analysis Status
ZONE_11	321.5	**Most Productive [cite: 18]**
ZONE_28	125.0	**Least Productive [cite: 18]**

Question 2: Recharge Method Costs

Comparative analysis of Digital vs. Manual recharge fees[cite: 20, 21].

```
-- Cost breakdown: Digital (0.61%) vs Manual (17.5 EGP) SELECT MONTH(c.payment_date) as Month, SUM(CASE WHEN recharge_method = 'UPG_DIGITAL_WALLET' THEN amount * 0.0061 ELSE 0 END) as Digital_Cost, SUM(CASE WHEN recharge_method = 'COLLECTION' AND device_type = 'pos' THEN 17.5 ELSE 0 END) as Manual_Cost FROM cash_ins c JOIN retailers r ON c.retailer_id = r.retailer_id GROUP BY MONTH(c.payment_date);
```

Cost Trend: App users are limited to digital wallets, while POS users can utilize both methods[cite: 22, 23, 26].

Final Strategic Recommendations

- **Re-allocate Resources:** Shift agents from low-performing zones (e.g., Zone 28) to high-pressure zones (e.g., Zone 11) to balance workload[cite: 19, 57].
- **Optimize Routes:** Implement route planning in low-productivity areas to reduce transit time for agents[cite: 57].
- **Incentivize Digital Adoption:** Convert POS users from Manual Collection (Fixed 17.5 EGP) to Digital (0.61%) to significantly reduce processing costs[cite: 26, 65].
- **Training Programs:** Provide additional training for agents in zones with high failure rates or slow completion times.